

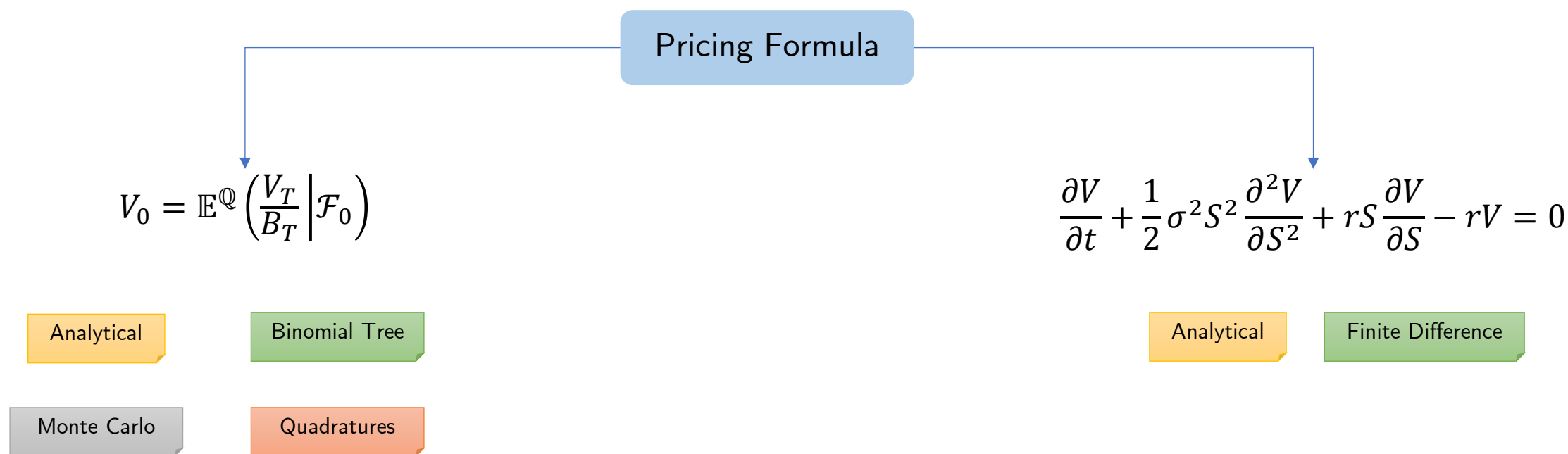
The Black-Scholes Model

(1. European Call Option Pricing)

Option Pricing Formula

Consider an equity derivative $V(S, t)$. There are two main pricing approaches

- ▶ Price via **Expectation**
- ▶ Price via solving a **PDE**





European Call Option (Analytical)

Consider a European call option having a payoff $V_T = (S_T - K)^+$

Let's calculate the analytical price of this call option at $t = 0$

$$\begin{aligned} V_0 &= \mathbb{E}^{\mathbb{Q}} \left(\frac{V_T}{B_T} \middle| \mathcal{F}_0 \right) = e^{-rT} \mathbb{E}^{\mathbb{Q}} [(S_T - K)^+] \\ &= e^{-rT} \mathbb{E}^{\mathbb{Q}} [(S_T - K) 1_{S_T > K}] \\ &= \underbrace{e^{-rT} \mathbb{E}^{\mathbb{Q}} [S_T 1_{S_T > K}]}_{\text{(II)}} - \underbrace{K e^{-rT} \mathbb{E}^{\mathbb{Q}} [1_{S_T > K}]}_{\text{(I)}} \end{aligned}$$

European Call Option (Analytical)

Let's evaluate (I)

$$\begin{aligned}Ke^{-rT} \mathbb{Q}(1_{S_T > K}) &= Ke^{-rT} \mathbb{Q}\left(S_0 \exp\left\{\left(r - \frac{1}{2}\sigma^2\right)T + \sigma\tilde{W}_T\right\} > K\right) \\&= Ke^{-rT} \mathbb{Q}\left(\exp\left\{\left(r - \frac{1}{2}\sigma^2\right)T + \sigma\tilde{W}_T\right\} > \frac{K}{S_0}\right) \\&= Ke^{-rT} \mathbb{Q}\left(\left(r - \frac{1}{2}\sigma^2\right)T + \sigma\tilde{W}_T > \ln\left(\frac{K}{S_0}\right)\right) \\&= Ke^{-rT} \mathbb{Q}\left(\tilde{W}_T > \frac{\ln\left(\frac{K}{S_0}\right) - \left(r - \frac{1}{2}\sigma^2\right)T}{\sigma}\right) \\&= Ke^{-rT} \mathbb{Q}\left(Z < \frac{\ln\left(\frac{S_0}{K}\right) + \left(r - \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}\right) \\&= Ke^{-rT} N(d_2)\end{aligned}$$

$\tilde{W}_T$ is a $\mathbb{Q}$ – Brownian

$$d_2 = \frac{\ln\left(\frac{S_0}{K}\right) + \left(r - \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}$$

European Call Option (Analytical)

Now Let's evaluate (II)

$$\begin{aligned}
 e^{-rT} \mathbb{E}^{\mathbb{Q}}[S_T 1_{S_T > K}] &= e^{-rT} \mathbb{E}^{\mathbb{Q}} \left[\left(S_0 \exp \left\{ \left(r - \frac{1}{2} \sigma^2 \right) T + \sigma \tilde{W}_T \right\} \right) \times 1_{S_T > K} \right] \\
 &= \mathbb{E}^{\mathbb{Q}} \left[\left(S_0 \exp \left\{ \left(-\frac{1}{2} \sigma^2 \right) T + \sigma \tilde{W}_T \right\} \right) \times 1_{S_T > K} \right] \\
 &= \mathbb{E}^{\mathbb{Q}^*} [(S_0) \times 1_{S_T > K}]
 \end{aligned}$$

$$\frac{d\mathbb{Q}^*}{d\mathbb{Q}} = e^{-\frac{1}{2}\sigma^2 T + \sigma \tilde{W}_T} \quad (\text{RN Derivative})$$

$$W_T^{\mathbb{Q}^*} = \tilde{W}_T - \sigma T \quad (\text{C.M.G})$$

The dynamics of S_T under stock-measure $\mathbb{Q}^*$ is given by

$$\begin{aligned}
 S_T &= S_0 \exp \left\{ \left(r - \frac{1}{2} \sigma^2 \right) T + \sigma (W_T^{\mathbb{Q}^*} + \sigma T) \right\} \\
 &= S_0 \exp \left\{ \left(r + \frac{1}{2} \sigma^2 \right) T + \sigma W_T^{\mathbb{Q}^*} \right\}
 \end{aligned}$$

Note : The drift of S_t is $(r + \sigma^2)$ instead of r under the stock measure

European Call Option (Analytical)

This leads to (II)

$$= \mathbb{E}^{\mathbb{Q}^*} [S_0 \times 1_{S_T > K}]$$

$$= S_0 \mathbb{E}^{\mathbb{Q}^*} [1_{S_T > K}]$$

Replace r by $(r + \sigma^2)$ in the previous derivation

$$= S_0 N(d_1) \quad d_1 = \frac{\ln\left(\frac{S_0}{K}\right) + \left(r + \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}$$

Price of the European Call Option

$$V_0 = S_0 N\left(\frac{\ln\left(\frac{S_0}{K}\right) + \left(r + \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}\right) - Ke^{-rT} N\left(\frac{\ln\left(\frac{S_0}{K}\right) + \left(r - \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}\right)$$